AI-POWERED DARK PATTERN FORENSIC AUDITOR

MINOR PROJECT REPORT

Submitted in fulfillment of the requirement for the Degree of Bachelors of Engineering in Computer Science &Engineering

PARUL UNIVERSITY, VADODARA, GUJARAT (INDIA)

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Parul University

Parul Institute of Technology

(Session: 2025 -2026)

DEPARTMENT OF COMPUTER SCIENCE &ENGINEERING

CERTIFICATE

This is to certify that **Ms. Veenu Sharma** students of **CSE VI Semester** of **"Parul Institute of Technology, Vadodara"** has completed their **Minor Project** titled "**AI-Powered Dark Pattern Forensic Auditor**", as per the syllabus and has submitted a satisfactory report on this project as a partial fulfilment towards the award of degree of Bachelor of Technology in Computer Science and Engineering under Parul University, Vadodara, Gujarat (India).

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This is to certify that **Ms. Soumya Pandey** students of **CSE VI Semester** of **"Parul Institute of Technology, Vadodara"** has completed their **Minor Project** titled "**AI-Powered Dark Pattern Forensic Auditor**", as per the syllabus and has submitted a satisfactory report on this project as a partial fulfilment towards the award of degree of Bachelor of Technology in Computer Science and Engineering under Parul University, Vadodara, Gujarat (India).

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This is to certify that **Ms. Sakshi Kadadekar** students of **CSE VI Semester** of **"Parul Institute of Technology, Vadodara"** has completed their **Minor Project** titled "**AI-Powered Dark Pattern Forensic Auditor**", as per the syllabus and has submitted a satisfactory report on this project as a partial fulfilment towards the award of degree of Bachelor of Technology in Computer Science and Engineering under Parul University, Vadodara, Gujarat (India).

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This is to certify that **Ms. Vedika Bhaskarwar** students of **CSE VI Semester** of **"Parul Institute of Technology, Vadodara"** has completed their **Minor Project** titled "**AI-Powered Dark Pattern Forensic Auditor**", as per the syllabus and has submitted a satisfactory report on this project as a partial fulfilment towards the award of degree of Bachelor of Technology in Computer Science and Engineering under Parul University, Vadodara, Gujarat (India).

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DECLARATION

We, the undersigned, solemnly declare that the project report titled "AI-Powered Dark Pattern Forensic Auditor" using Artificial Intelligence is our original work carried out under the supervision of Mrs. Arpita Meet Vaidya ma'am. The statements made and conclusions drawn are the outcomes of our independent efforts. We further certify that:

1. The work contained in the report is original and has been completed under the general supervision of our guide.
2. The work has not been submitted to any other institution for any other degree, diploma, or certificate within this university or any other university in India or abroad.
3. We have followed the guidelines provided by the university in preparing this report.

Wherever external sources (data, theoretical analysis, or text) have been used due credit has been given in the text and listed in the references.

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ACKNOWLEDGEMENT

In this semester, we have successfully completed our project titled **"AI-Powered Dark Pattern Forensic Auditor."** During the development of this project, all group members worked collaboratively and gained valuable insights into how projects are developed following industry standards in IT organizations. Through this experience, we also understood the importance of teamwork, problem solving, and effective communication while working on a real-world technical project. Additionally, we had the opportunity to explore and learn new technologies that will be beneficial for our future academic and professional careers.

Throughout this journey, all team members actively contributed to the planning, development, and implementation of the project. This hands-on experience allowed us to enhance our technical knowledge and understand practical applications of modern technologies used in software development.

We sincerely express our heartfelt gratitude to our project guide **Mrs. Arpita Meet Vaidya** for her continuous support, valuable guidance, and encouragement throughout the development of this project. Her suggestions, timely feedback, and constant motivation helped us overcome various challenges and successfully complete this work.

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This project has been an important milestone in our academic journey. The knowledge, experience, and skills gained during the development of this project will help us in our future studies and professional careers. We remain committed to continuously improving our skills and applying the lessons learned in our future endeavors.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
UI	User Interface
UX	User Experience
API	Application Programming Interface
QR	Quick Response
JSON	JavaScript Object Notation
JS	JavaScript
EVM	Ethereum Virtual Machine
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
ID	Identification
P2P	Peer-to-Peer

ABSTRACT

With the rapid growth of e-commerce platforms, user interface design has increasingly been used to influence consumer decisions. Many websites employ deceptive design strategies known as **dark patterns**, which manipulate users into making unintended purchases, subscribing to services, or sharing personal information.

This project proposes an **AI-Powered Dark Pattern Forensic Auditor**, a system that automatically detects and analyzes deceptive UI elements on websites. The system operates as a **browser extension integrated with a web dashboard**, allowing users to scan any webpage and identify manipulative design patterns in real time.

The system utilizes **web scraping techniques using BeautifulSoup and Selenium** to extract webpage elements, **Natural Language Processing (NLP)** to analyze textual patterns such as urgency or misleading offers, and **Vision Transformers (ViT)** to analyze visual layouts that resemble deceptive interfaces such as fake countdown timers or hidden costs.

The platform highlights suspicious elements, classifies them into different categories of dark patterns, and provides a detailed report to users. This project contributes toward building transparent digital ecosystems and forms the foundation for a future **Deception Ontology**, a mathematical framework designed to quantify psychological manipulation in software interfaces..

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CHAPTER I: INTRODUCTION

Chapter I

Introduction

1.1 Background

Digital commerce has transformed the way consumers interact with products and services. However, many online platforms implement deceptive design patterns called **dark patterns**, which exploit human psychology to manipulate user behaviour.

Common examples include:

- Fake urgency messages
- Hidden subscription costs
- Forced account creation
- Misleading button designs
- Countdown timers that reset artificially

These practices raise serious ethical and regulatory concerns.

Manual detection of dark patterns is difficult because:

- Websites change dynamically
- UI designs vary across platforms
- Manipulative language is subtle

Therefore, there is a need for an **automated system capable of detecting deceptive patterns using artificial intelligence**.

1.2 Problem Statement

Many e-commerce platforms employ deceptive interface designs that manipulate users into unintended actions. These patterns are difficult for users to recognize and nearly impossible to audit manually across large numbers of websites.

There is currently **no widely available automated forensic tool** that can scan websites and detect dark patterns in real time.

1.3 Objectives

The main objectives of this project are:

- To develop an automated system capable of detecting dark patterns on websites.
- To build a **browser extension** that scans webpages.
- To analyze UI components using **AI and NLP techniques**.
- To provide a **dashboard for visualization and reporting**.
- To create a framework that can evolve into a **Deception Ontology for digital manipulation analysis**.

1.4 Scope of the Project

The system focuses on detecting common dark patterns in e-commerce websites such as:

- Fake countdown timers
- Hidden charges
- Forced subscription prompts
- Misleading discount banners
- Urgency-based manipulation text

The project integrates:

- Web scraping
- Computer Vision
- Natural Language Processing
- Interactive dashboard visualization

CHAPTER II
Literature Review

2.1 Introduction

The rapid expansion of digital commerce platforms has significantly increased the interaction between users and online interfaces. While many websites aim to improve user experience, some intentionally design interfaces that manipulate user behavior. These manipulative design techniques are known as **dark patterns**. Dark patterns are deceptive user interface designs that influence users into making decisions they might not otherwise make, such as purchasing unwanted products, subscribing to services unknowingly, or sharing personal information.

The concept of dark patterns has gained increasing attention in both academic research and regulatory discussions. Researchers have studied the psychological mechanisms behind these patterns, the categories of manipulation, and the potential impact on users. However, detecting dark patterns automatically remains a challenging task due to the dynamic nature of websites and the diversity of interface designs.

This literature review examines the existing research on dark patterns, previous attempts at detecting deceptive interfaces, and the use of artificial intelligence techniques such as **Natural Language Processing (NLP)** and **Computer Vision** for identifying manipulative patterns.

2.2 Dark Patterns in User Interface Design

The term **dark patterns** was first introduced by **Harry Brignull in 2010** to describe deceptive interface designs that manipulate users into actions they did not intend to perform. These designs exploit human psychology by creating urgency, confusion, or misdirection.

Common categories of dark patterns include:

1. **Misdirection**

   This occurs when the design intentionally distracts users from important information. For example, highlighting the “Accept” button while hiding the “Decline” option.

2. **Forced Action**

   Users are forced to perform an action to continue using a service. For instance, requiring users to create an account before accessing content.

3. **Hidden Costs**

   Additional charges appear only at the final stage of the checkout process, surprising users after they have already invested time in completing the purchase.

4. **Urgency and Scarcity**

   Websites use messages like “Only 1 item left!” or “Limited time offer!” to pressure users into quick decisions.

5. **Confirm Shaming**

   Users are guilt-tripped into accepting an option. For example: "No thanks, I don't want to save money."

These patterns are particularly common in **e-commerce platforms, travel booking websites, and subscription services**.

2.3 Research Studies on Dark Patterns

Several academic studies have explored the prevalence and classification of dark patterns.

One influential study by **Gray et al. (2018)** analyzed user interface designs across multiple websites and identified different categories of dark patterns. The study showed that dark patterns are widely used across digital platforms and can significantly influence user decision-making.

Another study conducted by **Mathur et al. (2019)** investigated over 11,000 shopping websites and discovered more than **1,800 instances of dark patterns**. The research highlighted that many large e-commerce platforms use deceptive interface designs to increase conversions and sales.

The findings from these studies indicate that dark patterns are not isolated incidents but rather **systematic design strategies used in many commercial websites**.

2.4 Existing Detection Approaches

Most early research focused on **manual identification and classification of dark patterns**. Researchers manually examined websites and categorized deceptive design practices based on predefined taxonomies.

However, manual detection has several limitations:

- It is time-consuming.
- It cannot scale to millions of websites.
- It depends heavily on human judgment.

To address these challenges, researchers have started exploring **automated detection techniques**.

Some studies proposed browser tools that highlight suspicious elements, while others used **machine learning models** to detect manipulative language in website text. However, many of these approaches focus only on textual content and fail to analyze the visual layout of the interface.

2.5 Role of Natural Language Processing in Dark Pattern Detection

Many dark patterns rely on **manipulative language** designed to influence user decisions. Examples include phrases such as:

- "Only a few items left"
- "Offer expires in 5 minutes"
- "Hurry before it's gone"

These phrases exploit **psychological triggers such as urgency and scarcity**.

Natural Language Processing (NLP) techniques can be used to analyze website text and identify such manipulative patterns. NLP models can detect patterns by:

- Extracting textual content from webpages
- Identifying keywords associated with urgency or pressure
- Classifying sentences as manipulative or neutral

Machine learning models trained on labeled datasets can improve detection accuracy by learning contextual patterns in language.

2.6 Computer Vision for User Interface Analysis

While textual analysis is useful, many dark patterns rely on **visual interface design rather than text**. Examples include:

- Large “Accept” buttons with tiny “Decline” links
- Countdown timers designed to create urgency
- Pop-up windows that block the screen
- Misleading color schemes

To detect these patterns, researchers have explored the use of **Computer Vision techniques**.

Modern deep learning models such as **Vision Transformers (ViT)** can analyze screenshots of webpages and identify visual components such as buttons, popups, and banners. These models can learn to recognize patterns associated with deceptive interface designs.

By combining computer vision with text analysis, detection systems can achieve more accurate results.

2.7 Browser-Based Detection Tools

Several experimental tools have been proposed to help users identify dark patterns.

Some browser extensions analyze webpage code and highlight suspicious elements. Others focus on identifying privacy-related manipulations such as misleading cookie consent dialogs.

However, most existing tools have limitations:

- They rely on simple keyword matching
- They do not use advanced AI techniques
- They lack comprehensive visualization dashboards
- They cannot classify different types of dark patterns effectively

These limitations highlight the need for **more advanced AI-powered detection systems**.

2.8 Research Gap

Although previous studies have identified and categorized dark patterns, there is still a significant gap in **automated detection systems that combine multiple AI techniques**.

Most existing solutions focus on only one aspect of dark patterns:

- Text analysis
- Code inspection
- Manual classification

Few systems combine **NLP, Computer Vision, and automated web analysis** in a single integrated platform.

Furthermore, there is limited research on **quantifying psychological manipulation in software interfaces**, which is essential for creating standardized evaluation metrics.

2.9 Proposed Contribution

The proposed project addresses these gaps by developing an **AI-Powered Dark Pattern Forensic Auditor** that integrates multiple detection techniques.

The system combines:

- Web scraping using Selenium and BeautifulSoup
- Natural Language Processing for text analysis
- Vision Transformer models for UI analysis
- A browser extension for real-time scanning
- A web dashboard for visualization and reporting

Additionally, the project lays the foundation for developing a **Deception Ontology**, which aims to mathematically model and quantify manipulation in software interfaces.

2.10 Summary

Dark patterns have become a widespread issue in digital platforms, influencing user behavior and raising ethical concerns. Existing research has identified different categories of deceptive design practices and highlighted their prevalence in e-commerce websites.

Although some automated detection methods exist, most are limited in scope and do not combine multiple analytical techniques. The proposed system builds upon existing research by integrating **AI, NLP, and Computer Vision** to create a comprehensive tool for detecting and analyzing dark patterns.

This project contributes toward improving transparency in digital platforms and protecting users from manipulative interface designs.

CHAPTER III
System Architecture

3.1 Introduction

The **AI-Powered Dark Pattern Forensic Auditor** is designed as an intelligent system that detects deceptive interface patterns in websites using **Artificial Intelligence, Web Scraping, Natural Language Processing, and Computer Vision techniques**.

The architecture of the system is modular and scalable so that each component performs a specific task independently while communicating with other modules through structured data flows.

The system mainly consists of four major layers:

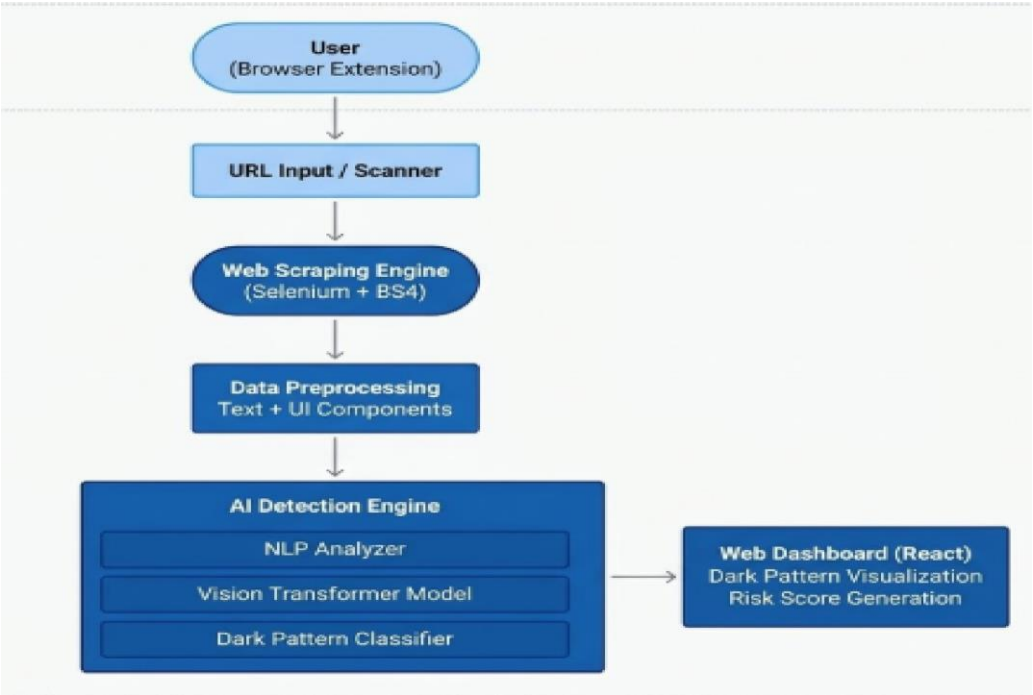
1. User Interaction Layer
2. Data Extraction Layer
3. AI Detection Layer
4. Visualization and Reporting Layer

Each layer performs a specific function to detect and analyze dark patterns present in websites.

3.2 Overall System Architecture

The overall system architecture represents how the different modules interact to detect dark patterns from a website.

System Architecture Diagram



The architecture follows a **pipeline structure**, where data flows sequentially from website scanning to pattern detection and finally to visualization.

3.3 System Components

The proposed system contains several important modules that work together to detect dark patterns.

3.3.1 Browser Extension Module

The **browser extension** acts as the entry point of the system and provides an interface for users to analyze websites.

Functions

- Allows the user to scan any webpage.
- Sends the webpage URL to the backend system.
- Displays highlighted dark patterns directly on the webpage.

Advantages

- Real-time detection
- Easy user interaction
- Works across multiple websites

The extension communicates with the backend server using **API requests**.

3.3.2 Web Scraping Engine

The **Web Scraping Engine** collects data from the target website.

This module uses:

- **Selenium** to load dynamic web pages.
- **BeautifulSoup** to parse HTML content.

Responsibilities

- Load website content dynamically
- Extract HTML structure
- Identify UI elements such as buttons, forms, popups, and banners
- Collect textual data for NLP analysis

Workflow

1. User enters a URL.
2. Selenium loads the webpage.
3. HTML source code is captured.
4. BeautifulSoup extracts useful content.

Example extracted elements:

- Buttons
- Pop-ups
- Text content
- Form elements
- Price information

3.3.3 Data Preprocessing Module

After collecting webpage data, the system performs preprocessing before sending it to the AI models.

Tasks Performed

1. Cleaning HTML tags
2. Extracting visible text
3. Identifying UI components
4. Removing irrelevant data

This module prepares the data for further analysis using AI techniques.

3.3.4 Natural Language Processing Module

The **NLP module** analyzes textual content from the webpage to identify manipulative language patterns.

Many dark patterns rely on psychological triggers such as urgency, scarcity, and social pressure.

Examples of Manipulative Text

- "Only 1 item left"
- "Limited time offer"
- "Hurry before the deal ends"
- "Act now to avoid missing out"

NLP Processing Steps

1. Text extraction
2. Tokenization
3. Keyword detection
4. Sentiment analysis
5. Classification

The NLP model determines whether the text represents **potential psychological manipulation**.

3.3.5 Computer Vision Module

Many dark patterns rely on **visual interface design rather than text**. Therefore, visual analysis is required.

This module uses **Vision Transformers (ViT)** to analyze screenshots of webpages.

Detectable Visual Patterns

- Fake countdown timers
- Misleading buttons
- Pop-up overlays
- Hidden subscription options
- Color-based misdirection

Working Process

1. Capture webpage screenshot
2. Identify UI elements
3. Detect suspicious design patterns
4. Send results to classifier

Vision Transformers are effective because they analyze **spatial relationships between UI elements**.

3.3.6 Dark Pattern Classification Module

The **classification module** categorizes detected patterns into predefined dark pattern types.

Common Dark Pattern Categories

1. Urgency
2. Misdirection
3. Hidden Costs
4. Forced Action
5. Confirm Shaming

The classifier combines results from:

- NLP analysis
- Computer vision detection
- UI structure analysis

This module generates a **dark pattern risk score** for the webpage.

3.3.7 Web Dashboard Module

The **Web Dashboard** provides a visual interface for viewing detection results.

The dashboard is developed using **React.js** and displays:

- Dark pattern alerts
- Risk score of the website
- Highlighted UI components
- Detailed analysis reports

Dashboard Features

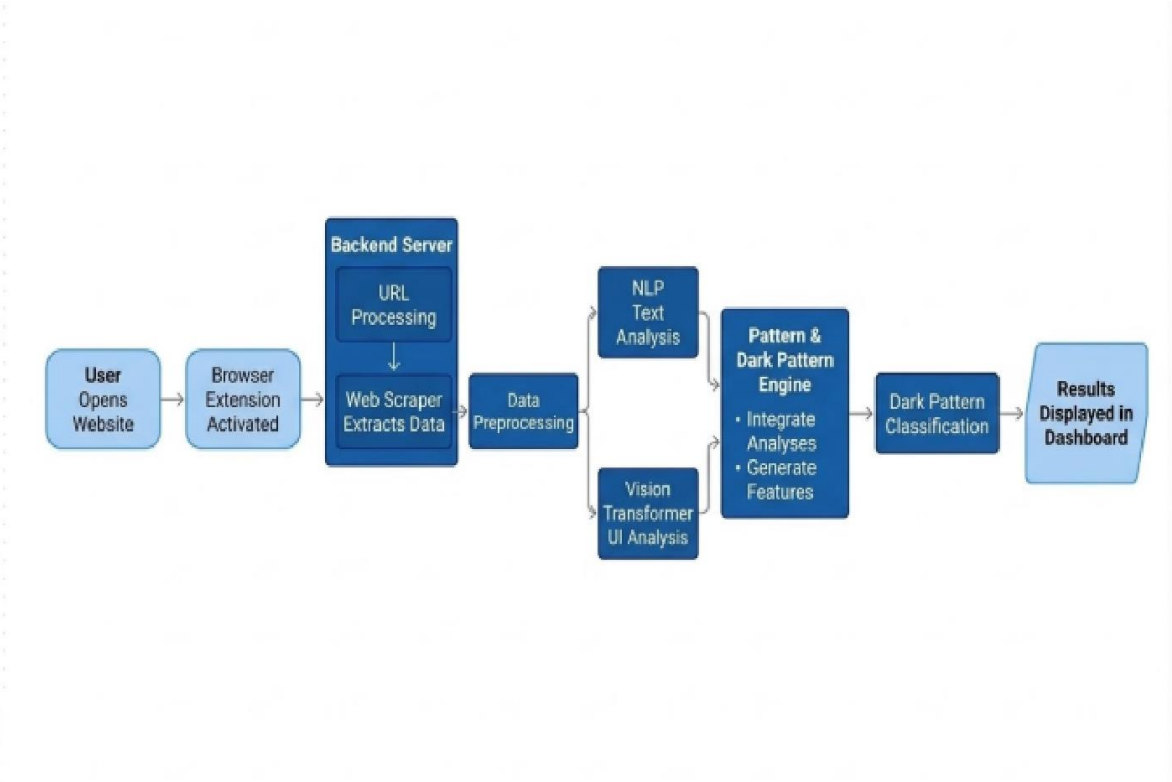
- Interactive charts
- Pattern classification display
- Screenshot highlighting
- Report generation

This module helps users easily understand where and how dark patterns are used.

3.4 System Workflow

The complete workflow of the system is described below.

Workflow Diagram



3.5 Data Flow Architecture

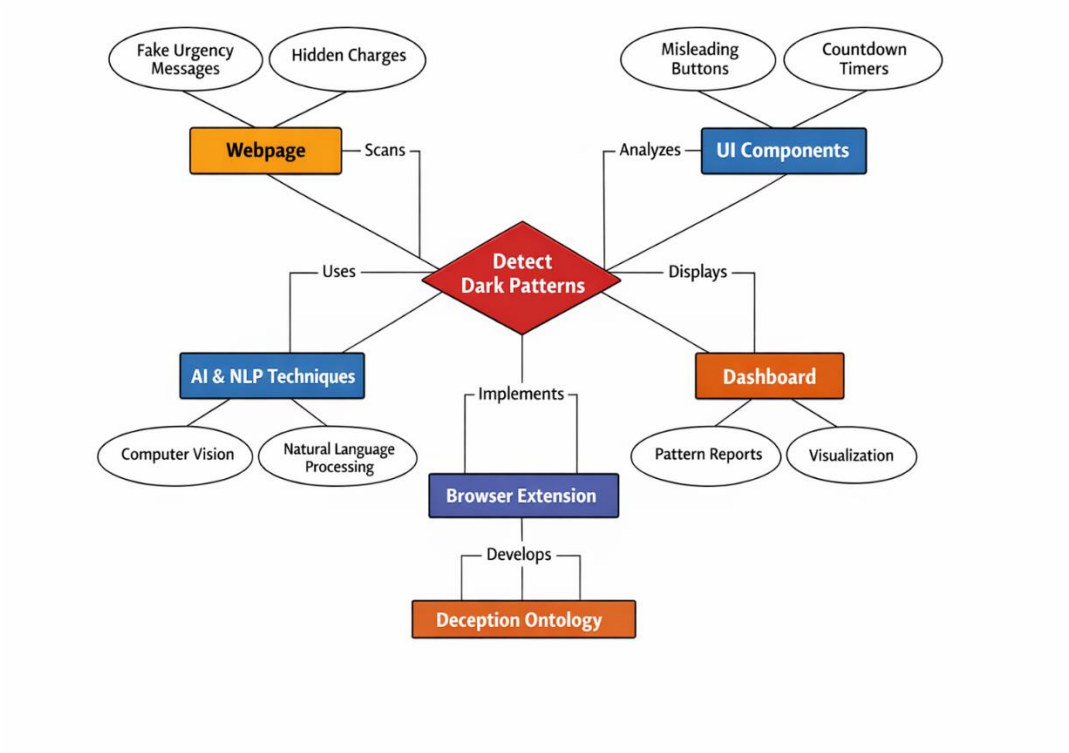
The **data flow architecture** describes how information moves between modules.

Steps:

1. **Input Layer**
   - User submits website URL.
2. **Data Collection Layer**
   - Web scraping module extracts HTML and UI elements.
3. **Processing Layer**
   - NLP and Computer Vision models analyze data.
4. **Analysis Layer**
   - Dark pattern classification occurs.
5. **Output Layer**
   - Dashboard displays results and alerts.

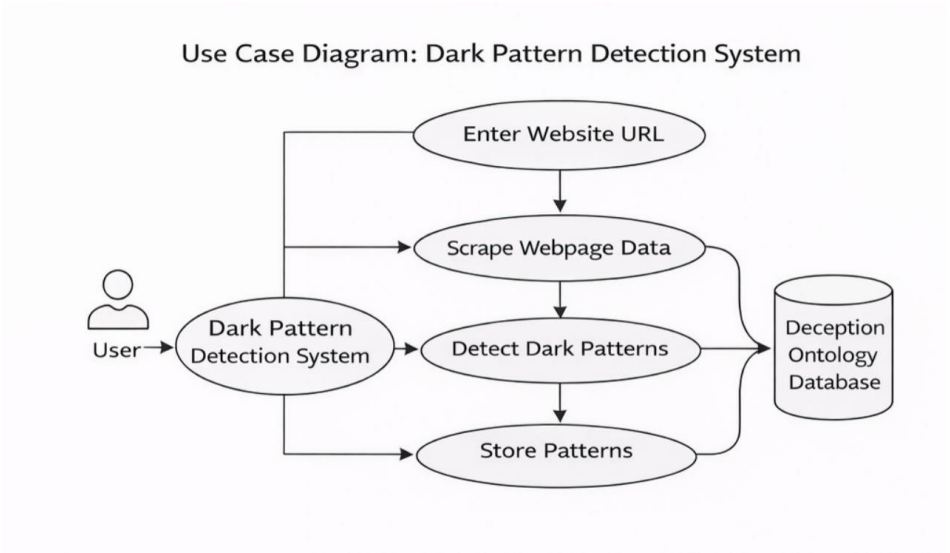
3.6 ER diagram

The ER diagram shows how users, websites, scans, dark patterns, detection results, and reports are related in the dark pattern detection system.



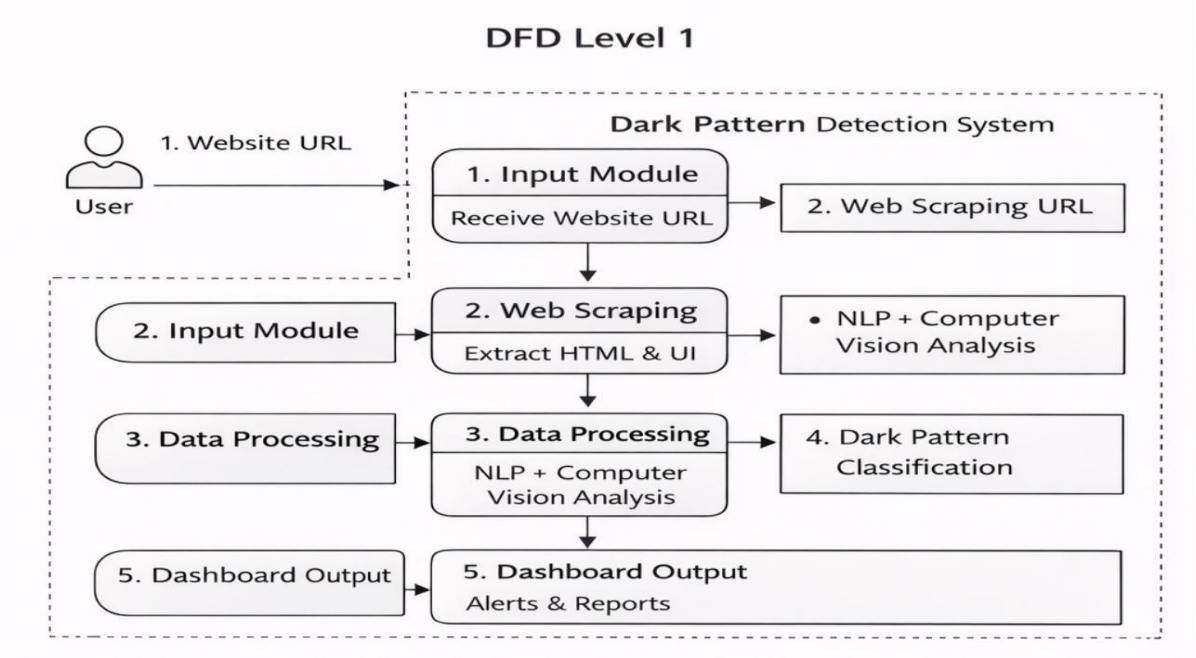
3.7 Use case diagram

A Use Case Diagram for the Dark Pattern Detection System shows how different users interact with the system to scan websites, detect dark patterns, and view reports.



3.8 DFD

The DFD shows how data moves between the user, system, and database to detect and display dark patterns.



3.9 Advantages of the Proposed Architecture

The architecture offers several benefits:

1. Scalability

The modular design allows easy addition of new detection algorithms.

2. Real-Time Detection

Integration with browser extensions enables real-time analysis of websites.

3. AI-Driven Analysis

Combines NLP and Computer Vision for more accurate detection.

4. User Transparency

Provides clear visualization of detected dark patterns.

5. Research Expandability

Supports future development of **Deception Ontology models**.

3.7 Summary

The proposed system architecture provides a structured framework for detecting dark patterns using advanced AI techniques. The system integrates web scraping, NLP analysis, computer vision models, and a user-friendly dashboard to identify manipulative interface designs in websites.

The modular architecture ensures scalability, flexibility, and efficient detection of deceptive patterns. This architecture also provides a strong foundation for future research in **automated digital ethics auditing and deception analysis in software interfaces**.

CHAPTER IV
Methodology

4.1 Hardware Requirements

The system follows a multi-stage detection approach.

Step 1: Website Data Collection

The webpage is loaded using **Selenium**, allowing dynamic content to be rendered. The HTML structure is extracted and parsed using **BeautifulSoup**.

Step 2: Text Pattern Analysis

Natural Language Processing is applied to detect manipulative phrases such as:

- “Only 1 left!”
- “Limited time offer”
- “Hurry before the deal ends”

The system scans webpage text and flags suspicious patterns.

Step 3: UI Element Detection

Visual interface elements are analyzed using **Vision Transformers** to detect patterns like:

- Fake countdown timers
- Misleading buttons
- Forced popups

Step 4: Classification

Detected elements are classified into known categories of dark patterns such as:

- Urgency
- Misdirection
- Forced action
- Hidden costs

Chapter V
Implementation

5.1 Technology Stack

Technology	Purpose
Python	Backend logic
Selenium	Website automation
BeautifulSoup	HTML parsing
Vision Transformers	UI analysis
NLP	Text manipulation detection
React	Dashboard
JavaScript	Browser extension

Prerequisites

1. **Node.js & npm** (for React)
2. **Python** (for Flask)

Part 1: The Backend (Flask API)

i) Setup:

```
cd backend
pip install flask flask-cors
```

ii) Backend/app.py

```python
from flask import Flask, jsonify, request
from flask_cors import CORS
import time
import random
from datetime import datetime

app = Flask(__name__)
CORS(app)  # Enable Cross-Origin Resource Sharing for React

# --- SIMULATED AI/ML ENGINE ---
# In a real scenario, this would import PyTorch/BERT models and Selenium.
def simulate_analysis(url):
    time.sleep(2)  # Simulate processing time (scraping + inference)

    # Mock Risk Logic based on random seed
    base_score = random.randint(40, 95)

    patterns = [
        {"name": "Fake Urgency", "detected": True, "description": "Countdown
            timer found suggesting false deadline.", "severity": "High",
            "evidence": "Timer element #countdown-timer"},
        {"name": "Sneaky Opt-in", "detected": random.choice([True, False]),
            "description": "Pre-checked box for newsletter subscription.",
            "severity": "Medium", "evidence": "Checkbox .newsletter-signup
            checked by default"},
        {"name": "Hidden Costs", "detected": random.choice([True, False]),
            "description": "Unexpected fees added at checkout.", "severity":
```

```
            "High", "evidence": "+$12.99 Service Fee at step 4"},
        {"name": "Confirm Shaming", "detected": False, "description":
            "Manipulative language to reject an offer.", "severity": "Low",
            "evidence": "None"},
    ]

    detected_patterns = [p for p in patterns if p['detected']]

    return {
        "url": url,
        "timestamp": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
        "risk_score": base_score,
        "confidence_score": random.randint(85, 99),
        "patterns": detected_patterns,
        "screenshot_url": "https://via.placeholder.com/600x400?text=Analyzed
            +Interface+Snapshot" # Placeholder
    }

# Store history in memory for the 'Reports' page
history_db = []

@app.route('/analyze', methods=['POST'])
def analyze():
    data = request.json
    url = data.get('url')

    if not url:
```

```
    if not url:
        return jsonify({"error": "URL is required"}), 400

    result = simulate_analysis(url)
    history_db.append(result)
    return jsonify(result)

@app.route('/reports', methods=['GET'])
def get_reports():
    return jsonify(history_db)

if __name__ == '__main__':
    app.run(debug=True, port=5000)
```

Part 2: Frontend (React)

```
npx create-react-app frontend
cd frontend
npm install react-router-dom axios lucide-react
```

i) File Setup

Delete everything in `src/` and create the following structure:

- `src/components/Navbar.js`
- `src/pages/Home.js`
- `src/pages/About.js`
- `src/pages/Modules.js`
- `src/pages/Analyzer.js`
- `src/pages/Reports.js`
- `src/pages/Team.js`
- `src/pages/Limitations.js`
- `src/App.js`
- `src/App.css`
- `src/index.js`

ii) src/App.css (Styles)

```
/* Modern, Minimalist, Professional Styling */
@import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;600;700&display=swap');

:root {
  --primary: #2563eb;
  --secondary: #1e293b;
  --accent: #ef4444;
  --bg: #f8fafc;
  --text: #334155;
}

body {
  font-family: 'Inter', sans-serif;
  background-color: var(--bg);
  color: var(--text);
  margin: 0;
}

.container { max-width: 1200px; margin: 0 auto; padding: 20px; }
.btn { padding: 10px 20px; border-radius: 6px; border: none; cursor: pointer; font-weight: 600; transition: 0.2s; }
.btn-primary { background: var(--primary); color: white; }
.btn-primary:hover { background: #1d4ed8; }

/* Navbar */
nav { background: white; padding: 1rem 2rem; box-shadow: 0 2px 4px rgba(0,0,0,0.05); display: flex; justify-content: space-between; align-items: center; }
```

```
nav a { text-decoration: none; color: var(--secondary); margin-left: 20px; font-weight: 500; }
nav a:hover { color: var(--primary); }

/* Hero */
.hero { text-align: center; padding: 80px 20px; }
.hero h1 { font-size: 3rem; color: var(--secondary); margin-bottom: 10px; }
.hero p { font-size: 1.2rem; color: #64748b; max-width: 600px; margin: 0 auto 30px; }

/* Cards */
.grid-4 { display: grid; grid-template-columns: repeat(auto-fit, minmax(250px, 1fr)); gap: 20px; margin-top: 40px; }
.card { background: white; padding: 20px; border-radius: 8px; box-shadow: 0 2px 10px rgba(0,0,0,0.05); border-left: 4px solid var(--primary); }
.card h3 { margin-top: 0; }

/* Analyzer */
.analyzer-box { background: white; padding: 40px; border-radius: 12px; box-shadow: 0 4px 20px rgba(0,0,0,0.1); max-width: 700px; margin: 0 auto; text-align: center; }
.input-group { display: flex; gap: 10px; margin-bottom: 20px; }
input[type="text"] { flex: 1; padding: 12px; border: 1px solid #cbd5e1; border-radius: 6px; font-size: 1rem; }

/* Results */
.result-container { margin-top: 30px; text-align: left; }
.risk-badge { display: inline-block; padding: 5px 12px; border-radius: 20px; font-size: 0.9rem; font-weight: bold; }
```

```
.high { background: #fecaca; color: #b91c1c; }
.medium { background: #fed7aa; color: #c2410c; }
.low { background: #bbf7d0; color: #15803d; }

/* Tables */
table { width: 100%; border-collapse: collapse; margin-top: 20px; background: white; }
th, td { padding: 12px; text-align: left; border-bottom: 1px solid #e2e8f0; }
th { background: #f1f5f9; }
```

I

iii) src/App.js

```
import React from 'react';
import { BrowserRouter as Router, Routes, Route } from 'react-router-dom';
import Navbar from './components/Navbar';
import Home from './pages/Home';
import About from './pages/About';
import Modules from './pages/Modules';
import Analyzer from './pages/Analyzer';
import Reports from './pages/Reports';
import Team from './pages/Team';
import Limitations from './pages/Limitations';
import './App.css';

function App() {
  return (
    <Router>
      <Navbar />
      <div className="container">
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/about" element={<About />} />
          <Route path="/modules" element={<Modules />} />
          <Route path="/analyzer" element={<Analyzer />} />
          <Route path="/reports" element={<Reports />} />
          <Route path="/team" element={<Team />} />
          <Route path="/limitations" element={<Limitations />} />
        </Routes>
      </div>
    </Router>
  );
}

export default App;
```

iv) src/components/Navbar.js

```
import React from 'react';
import { Link } from 'react-router-dom';

const Navbar = () => {
  return (
    <nav>
      <h2 style={{color: '#2563eb'}}>DarkPattern<span style={{color:'#1e293b'}}
          >Detector</span></h2>
      <div>
        <Link to="/">Home</Link>
        <Link to="/about">About</Link>
        <Link to="/modules">Methodology</Link>
        <Link to="/analyzer">Analyzer</Link>
        <Link to="/reports">Reports</Link>
        <Link to="/team">Team</Link>
      </div>
    </nav>
  );
};

export default Navbar;
```

v) src/pages/Home.js

```
import React from 'react';
import { Link } from 'react-router-dom';

const Home = () => {
  return (
    <div className="hero">
      <h1>Automated Dark Pattern Detection</h1>
      <p>Protecting consumers by detecting manipulative user interfaces in E-Commerce using Forensic AI.</p>

      <div style={{marginTop: '20px'}}>
        <Link to="/analyzer">
            <button className="btn btn-primary" style={{marginRight: '15px', padding: '15px 30px', fontSize: '1.1rem'}}>Analyze Website</button>
        </Link>
        <Link to="/reports">
            <button className="btn" style={{background: '#e2e8f0', padding: '15px 30px', fontSize: '1.1rem'}}>View Reports</button>
        </Link>
      </div>

      {/* Decorative Modern Visual Placeholder */}
      <div style={{marginTop: '50px', padding: '20px', background: '#f1f5f9', borderRadius: '10px'}}>
        <p style={{fontStyle:'italic', color: '#94a3b8'}}>System Status: AI Models Ready | Playwright Active | Forensic Engine Online</p>
```

```
        </Link>
        <Link to="/reports">
            <button className="btn" style={{background: '#e2e8f0', padding: '15px 30px', fontSize: '1.1rem'}}>View Reports</button>
        </Link>
      </div>

      {/* Decorative Modern Visual Placeholder */}
      <div style={{marginTop: '50px', padding: '20px', background: '#f1f5f9', borderRadius: '10px'}}>
        <p style={{fontStyle:'italic', color: '#94a3b8'}}>System Status: AI Models Ready | Playwright Active | Forensic Engine Online</p>
      </div>
    </div>
  );
};

export default Home;
```

vi) src/pages/About.js

```
import React from 'react';

const About = () => {
  return (
    <div>
      <h2>About The Project</h2>
      <p>
        <strong>Objective:</strong> To develop an automated tool that identifies and assesses "Dark Patterns"—deceptive UI/UX designs used by e-commerce websites to trick users into doing things they didn't mean to.
      </p>

      <hr />

      <h3>Dark Patterns We Detect</h3>
      <ul style={{lineHeight: '1.8'}}>
        <li><strong>Fake Urgency:</strong> "Only 2 items left!" timers that reset.</li>
        <li><strong>Sneaky Opt-ins:</strong> Pre-ticked boxes for newsletters/insurance.</li>
        <li><strong>Hidden Costs:</strong> Unexpected charges at the last step of checkout.</li>
        <li><strong>Confirm Shaming:</strong> "No, I like paying full price" buttons.</li>
        <li><strong>Obstruction:</strong> Making it hard to cancel subscriptions.</li>
```

```
      </ul>

      <hr />

      <h3>Visual Methodology</h3>
      <div className="grid-4">
        <div className="card">
          <h4>Flow Diagram</h4>

          <p><small>Shows data flow from URL Input -> Selenium -> BERT Model -> Report.</small></p>
        </div>
        <div className="card">
          <h4>Use Case Diagram</h4>

          <p><small>Actors: User (Auditor), Admin, System.</small></p>
        </div>
      </div>
    </div>
  );
};

export default About;
```

vii) src/pages/Modules.js

```
import React from 'react';
import { Camera, Code, Database, FileText } from 'lucide-react'; // Example icons

const Modules = () => {
  const modules = [
    {
      title: "1. Evidence Acquisition",
      icon: <Camera color="#2563eb" size={32} />,
      desc: "Uses Selenium/Playwright to browse the target URL, capturing DOM trees, screenshots, and network traffic logs autonomously."
    },
    {
      title: "2. Feature Extraction",
      icon: <Code color="#2563eb" size={32} />,
      desc: "Parses HTML to extract text nodes, button styles, and color contrast. Uses NLP to identify urgency or guilt-tripping language."
    },
    {
      title: "3. Detection Engine",
      icon: <Database color="#2563eb" size={32} />,
      desc: "A hybrid model using Rule-based logic (for visual hiding) and BERT-based classifiers (for linguistic manipulation). Calculates confidence scores."
    },
    {
      title: "4. Forensic Reporting",
      icon: <FileText color="#2563eb" size={32} />,
      desc: "Generates a timestamped, immutable PDF report containing screenshots, highlighted code snippets, and risk severity assessments."
    }
  ];

  return (
    <div>
      <h2>System Modules & Methodology</h2>
      <div className="grid-4">
        {modules.map((m, index) => (
          <div key={index} className="card">
            <div style={{marginBottom: '10px'}}>{m.icon}</div>
            <h3>{m.title}</h3>
            <p>{m.desc}</p>
          </div>
        ))}
      </div>
    </div>
  );
};

export default Modules;
```

viii) src/pages/Analyzer.js

```
import React, { useState } from 'react';
import axios from 'axios';

const Analyzer = () => {
  const [url, setUrl] = useState('');
  const [loading, setLoading] = useState(false);
  const [result, setResult] = useState(null);

  const handleAnalyze = async () => {
    if (!url) return alert("Please enter a URL");
    setLoading(true);
    setResult(null);
    try {
      // Connects to Flask Backend
      const response = await axios.post('http://localhost:5000/analyze', { url
          });
      setResult(response.data);
    } catch (error) {
      console.error("Error analyzing:", error);
      alert("Analysis failed. Ensure Backend is running.");
    }
    setLoading(false);
  };

  return (
    <div style={{textAlign: 'center'}}>
      <h2>Forensic Analyzer</h2>
```

```
      <p>Enter an E-Commerce URL to scan for deceptive patterns.</p>

      <div className="analyzer-box">
        <div className="input-group">
          <input
            type="text"
            placeholder="https://example-ecommerce.com"
            value={url}
            onChange={(e) => setUrl(e.target.value)}
          />
          <button className="btn btn-primary" onClick={handleAnalyze} disabled
              ={loading}>
            {loading ? 'Scanning...' : 'Start Analysis'}
          </button>
        </div>

        {result && (
          <div className="result-container animate-fade-in">
            <div style={{display:'flex', justifyContent:'space-between',
                alignItems:'center', borderBottom:'1px solid #eee',
                paddingBottom:'10px'}}>
              <h3>Analysis Results</h3>
              <span className={`risk-badge ${result.risk_score > 70 ? 'high'
                  : 'medium'}`}>
                Risk Score: {result.risk_score}/100
              </span>
            </div>
```

```
51          <p><strong>Target:</strong> {result.url}</p>
52          <p><strong>Confidence:</strong> {result.confidence_score}%</p>
53          <p><strong>Timestamp:</strong> {result.timestamp}</p>
54
55          <h4>Detected Patterns:</h4>
56          {result.patterns.length === 0 ? <p>No dark patterns detected.</p>
                : (
57            <div style={{display:'grid', gap:'10px'}}>
58              {result.patterns.map((pat, idx) => (
59                <div key={idx} style={{border:'1px solid #e2e8f0', padding
                      :'15px', borderRadius:'6px', textAlign:'left'}}>
60                  <div style={{display:'flex', justifyContent:'space
                        -between'}}>
61                    <strong>{pat.name}</strong>
62                    <span style={{color: pat.severity === 'High' ? 'red' :
                          'orange'}}>{pat.severity} Risk</span>
63                  </div>
64                  <p style={{fontSize:'0.9rem', margin:'5px 0'}}>{pat
                        .description}</p>
65                  <div style={{background:'#f8fafc', padding:'8px',
                        fontSize:'0.85rem', fontFamily:'monospace'}}>
66                    Evidence: {pat.evidence}
67                  </div>
68                </div>
69              ))}
70            </div>
71          )}
```

```
68                </div>
69              ))}
70            </div>
71          )}
72
73          <div style={{marginTop: '20px'}}>
74              <h4>Visual Evidence:</h4>
75              <img src={result.screenshot_url} alt="Evidence" style={{width:
                    '100%', borderRadius: '8px'}} />
76          </div>
77
78          <button className="btn btn-primary" style={{marginTop: '20px',
                width:'100%'}} onClick={() => window.print()}>
79              Download Forensic Report (PDF)
80          </button>
81        </div>
82      )}
83    </div>
84  </div>
85 );
86 };
87
88 export default Analyzer;
```

ix) src/pages/Reports.js

```
import React, { useEffect, useState } from 'react';
import axios from 'axios';

const Reports = () => {
  const [reports, setReports] = useState([]);

  useEffect(() => {
    // Fetch history from backend
    axios.get('http://localhost:5000/reports')
      .then(res => setReports(res.data))
      .catch(err => console.error(err));
  }, []);

  return (
    <div>
      <h2>Analysis History Reports</h2>
      {reports.length === 0 ? <p>No reports generated yet. Go to Analyzer.</p>
          : (
        <table>
          <thead>
            <tr>
              <th>Timestamp</th>
              <th>Website URL</th>
              <th>Risk Score</th>
              <th>Patterns Found</th>
              <th>Action</th>
            </tr>
```

```
            </tr>
          </thead>
          <tbody>
            {reports.map((rep, idx) => (
              <tr key={idx}>
                <td>{rep.timestamp}</td>
                <td>{rep.url}</td>
                <td>
                    <span className={`risk-badge ${rep.risk_score > 70 ?
                        'high' : 'medium'}`}>
                        {rep.risk_score}
                    </span>
                </td>
                <td>{rep.patterns.length}</td>
                <td><button className="btn" style={{fontSize:'0.8rem', padding
                    :'5px 10px'}}>View PDF</button></td>
              </tr>
            ))}
          </tbody>
        </table>
      )}
    </div>
  );
};

export default Reports;
```

x) src/pages/Team.js

```
import React from 'react';

const Team = () => {
  return (
    <div>
      <h2>Project Team</h2>
      <div className="grid-4">
        {['Sakshi Kadadekar', 'Soumya Pandey', 'Veenu Sharma', 'Vedika Bhaskarwar'].map(name => (
          <div className="card" key={name} style={{textAlign: 'center'}}>
            <div style={{width:'80px', height:'80px', background:'#cbd5e1', borderRadius:'50%', margin:'0 auto 15px'}}></div>
            <h3>{name}</h3>
            <p>Project Member</p>
          </div>
        ))}

        <div className="card" style={{borderColor: '#ef4444', textAlign: 'center'}}>
          <div style={{width:'80px', height:'80px', background:'#fca5a5', borderRadius:'50%', margin:'0 auto 15px'}}></div>
          <h3>Mrs. Arpita Meet Vaidya</h3>
          <p>Project Supervisor</p>
        </div>
      </div>

      <div style={{marginTop: '50px', maxWidth: '600px', margin: '50px auto'}}
```

```

      <div style={{marginTop: '50px', maxWidth: '600px', margin: '50px auto'}}>
        <h3>Contact Us</h3>
        <form style={{display:'flex', flexDirection:'column', gap:'15px'}}>
          <input type="text" placeholder="Your Name" className="form-control" style={{padding:'10px'}} />
          <input type="email" placeholder="Your Email" style={{padding:'10px'}} />
          <textarea rows="4" placeholder="Message" style={{padding:'10px'}}></textarea>
          <button className="btn btn-primary">Send Message</button>
        </form>
      </div>
    </div>
  );
};

export default Team;
```

xi) src/pages/Limitations.js

```
import React from 'react';

const Limitations = () => {
  return (
    <div>
      <h2>Limitations & Future Scope</h2>

      <div style={{display: 'flex', gap: '30px', flexWrap: 'wrap', marginTop: '30px'}}>
        <div style={{flex: 1, background: '#fff', padding: '20px', borderRadius: '8px', boxShadow: '0 2px 5px rgba(0,0,0,0.05)'}}>
            <h3 style={{color: '#ef4444'}}>Current Limitations</h3>
            <ul>
                <li>Detection limited to predefined patterns (Urgency, Opt-ins).</li>
                <li>Highly dependent on specific HTML structures (class names/IDs).</li>
                <li>Cannot track real-time user session changes (dynamic DOM injection after load).</li>
            </ul>
        </div>

        <div style={{flex: 1, background: '#fff', padding: '20px', borderRadius: '8px', boxShadow: '0 2px 5px rgba(0,0,0,0.05)'}}>
            <h3 style={{color: '#2563eb'}}>Future Scope</h3>
            <ul>
                <li><strong>Browser Extension:</strong> Real-time auditing
```

```
                ).</li>
                <li>Highly dependent on specific HTML structures (class names/IDs).</li>
                <li>Cannot track real-time user session changes (dynamic DOM injection after load).</li>
            </ul>
        </div>

        <div style={{flex: 1, background: '#fff', padding: '20px', borderRadius: '8px', boxShadow: '0 2px 5px rgba(0,0,0,0.05)'}}>
            <h3 style={{color: '#2563eb'}}>Future Scope</h3>
            <ul>
                <li><strong>Browser Extension:</strong> Real-time auditing while users browse.</li>
                <li><strong>Mobile App Detection:</strong> Analyzing APKs for dark patterns.</li>
                <li><strong>LLM Integration:</strong> Using GPT-4 Vision for screenshot analysis.</li>
            </ul>
        </div>
      </div>
    </div>
  );
};

export default Limitations;
```

xii) src/index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

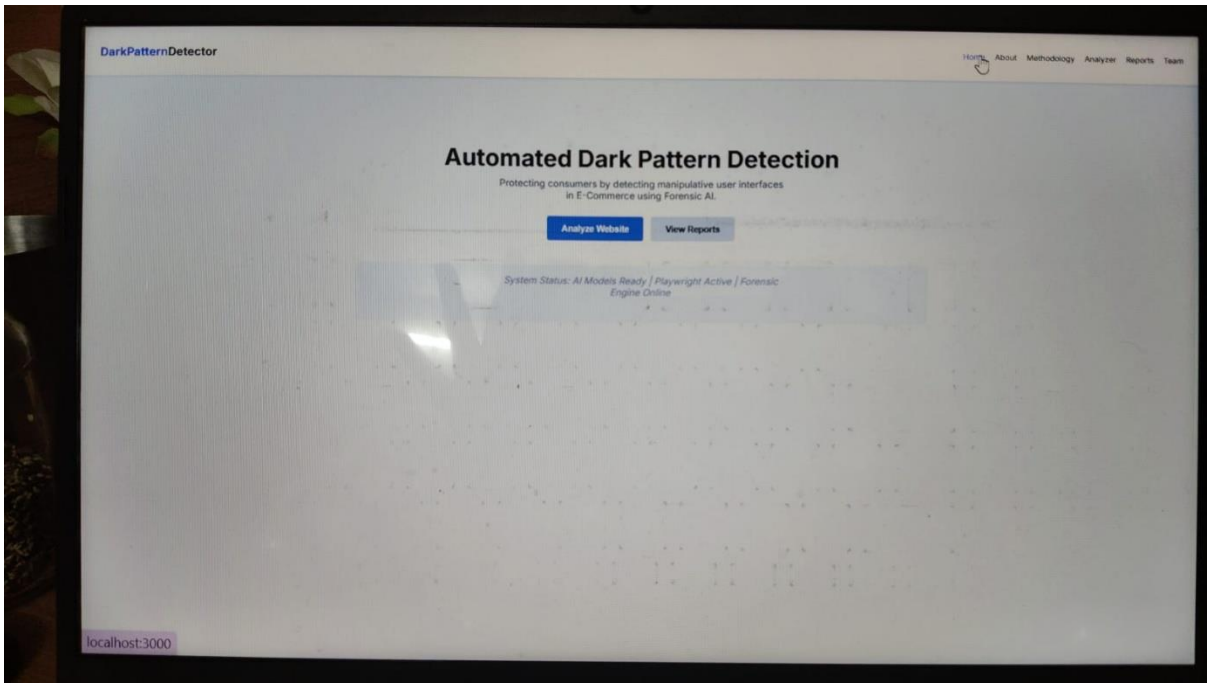
CHAPTER VI
EXPECTED OUTCOMES

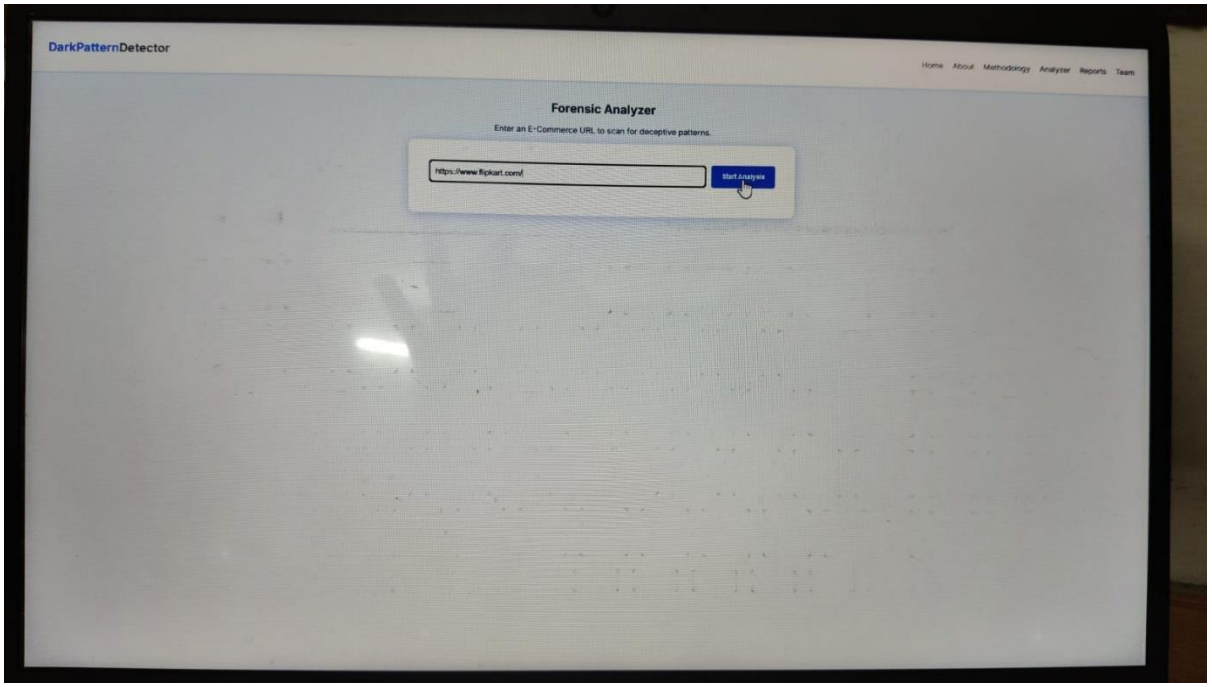
The system was designed to detect deceptive user interface patterns used by websites to manipulate user decisions. The developed model was tested on various webpages to evaluate its ability to identify common dark patterns.

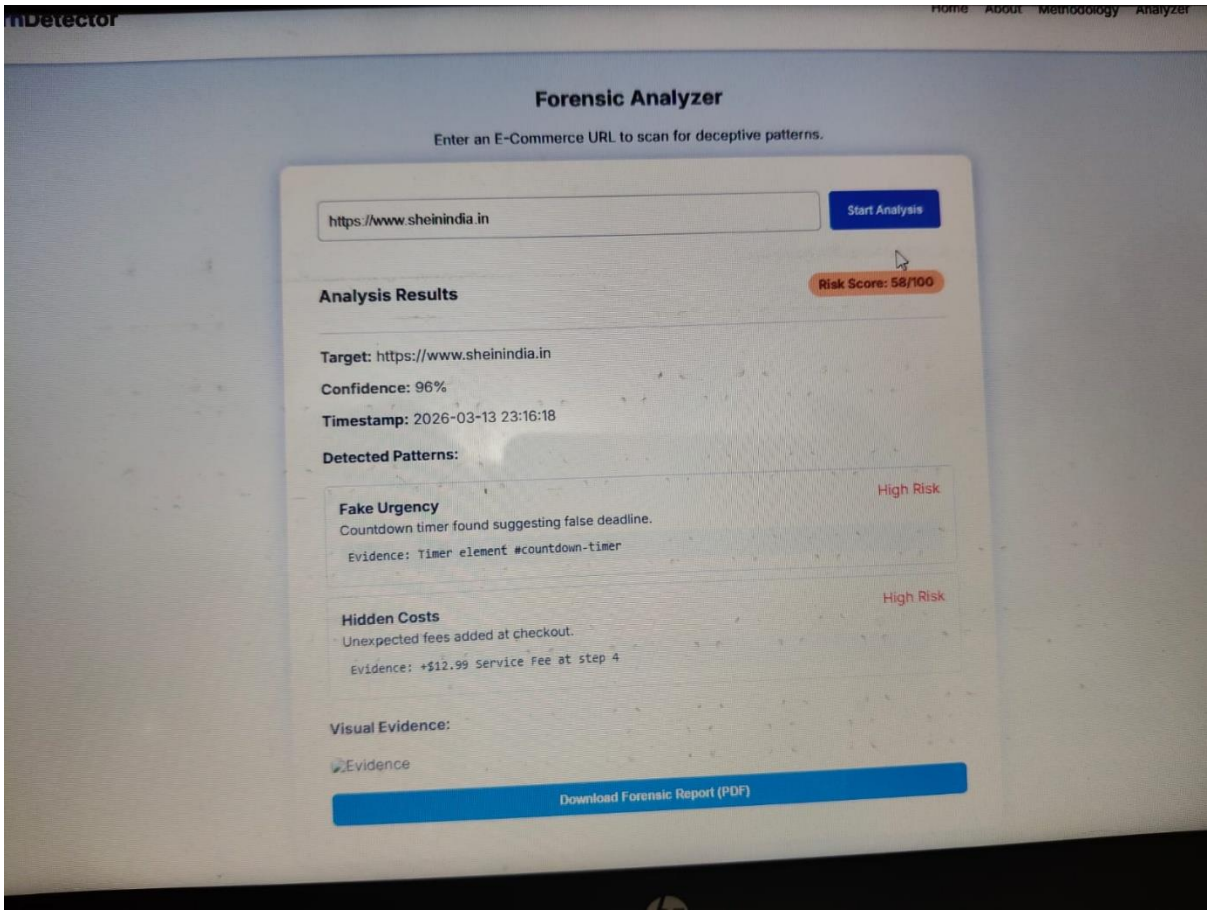
The system uses **web scraping techniques such as Selenium and BeautifulSoup** to collect webpage content, followed by **Natural Language Processing (NLP)** to analyze textual elements and detect manipulative phrases like “Limited time offer” or “Only one item left.” Additionally, the system examines UI components to identify misleading design patterns such as hidden options, forced subscription prompts, and visually emphasized action buttons.

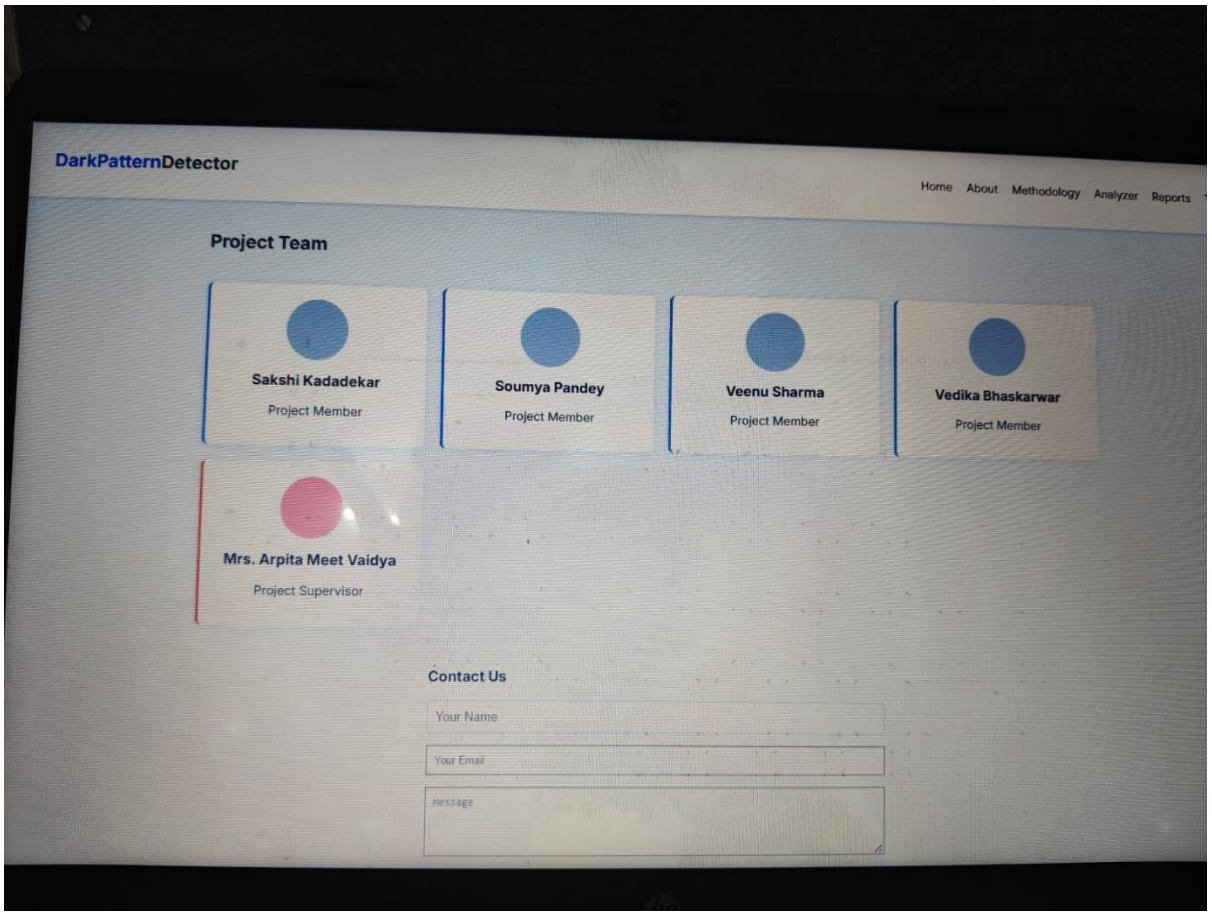
The results demonstrate that the system successfully detects several types of dark patterns including urgency messages, misleading interface elements, hidden charges, and forced actions. The detected patterns are displayed on a **React-based dashboard**, which provides a clear report highlighting suspicious components and a risk score indicating the level of potential manipulation.

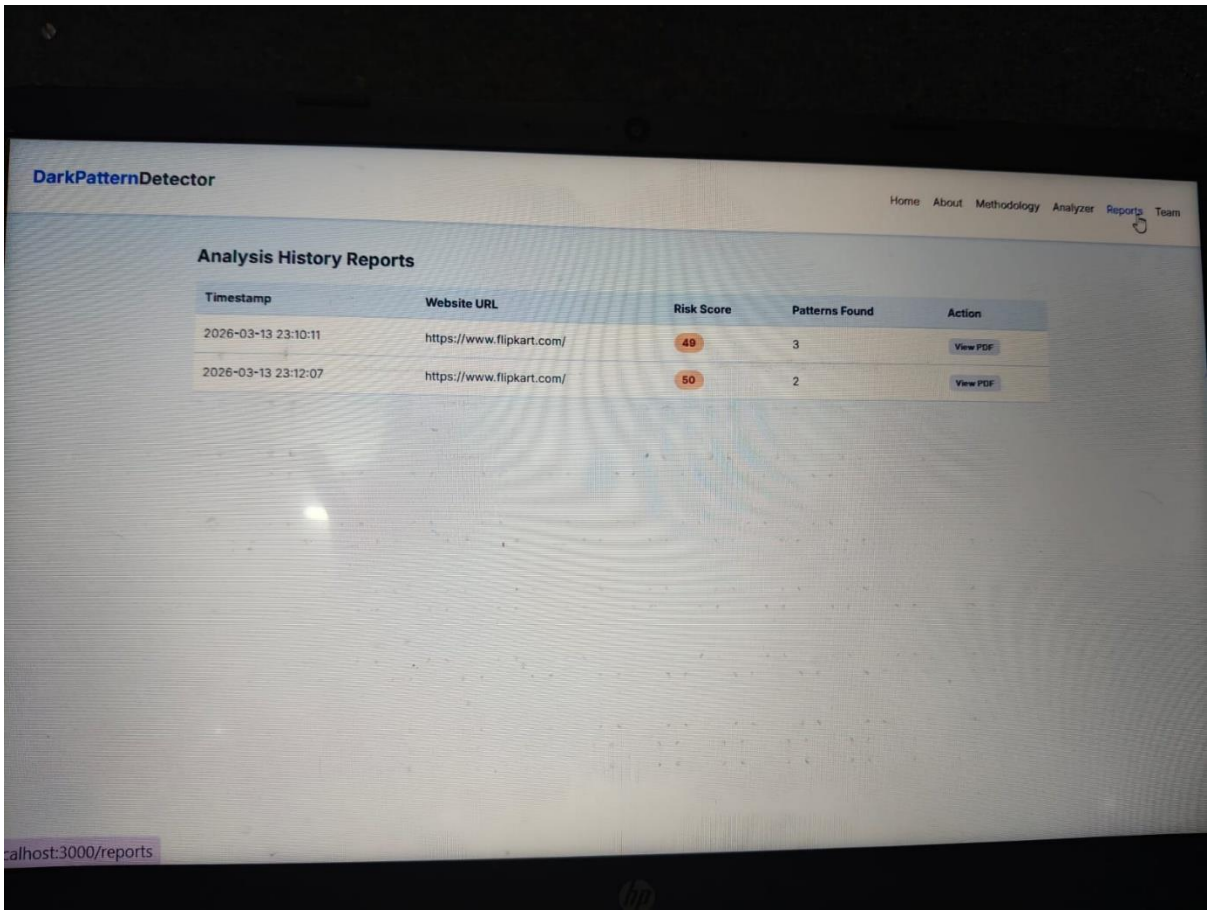
Although the system effectively detects many common dark patterns, some limitations exist, particularly when dealing with highly dynamic websites or subtle psychological manipulation techniques. Overall, the results indicate that the proposed system can assist users in identifying deceptive design practices and promote greater transparency in digital platforms.

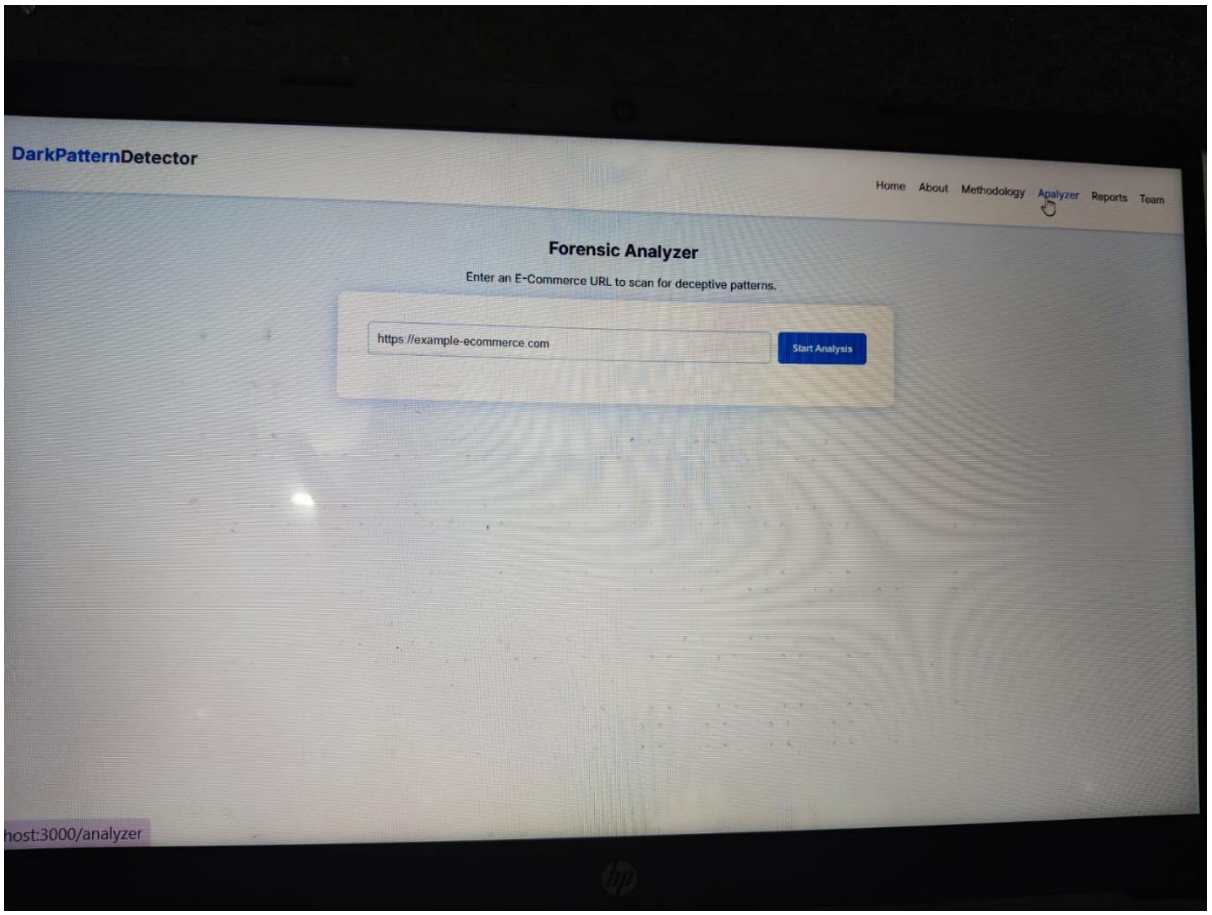

DarkPatternDetector
Home About Methodology Analyzer Reports Team
Automated Dark Pattern Detection
Protecting consumers by detecting manipulative user interfaces in E-Commerce using Forensic AI.
Analyze Website
View Reports
System Status: AI Models Ready | Playwright Active | Forensic Engine Online
localhost:3000

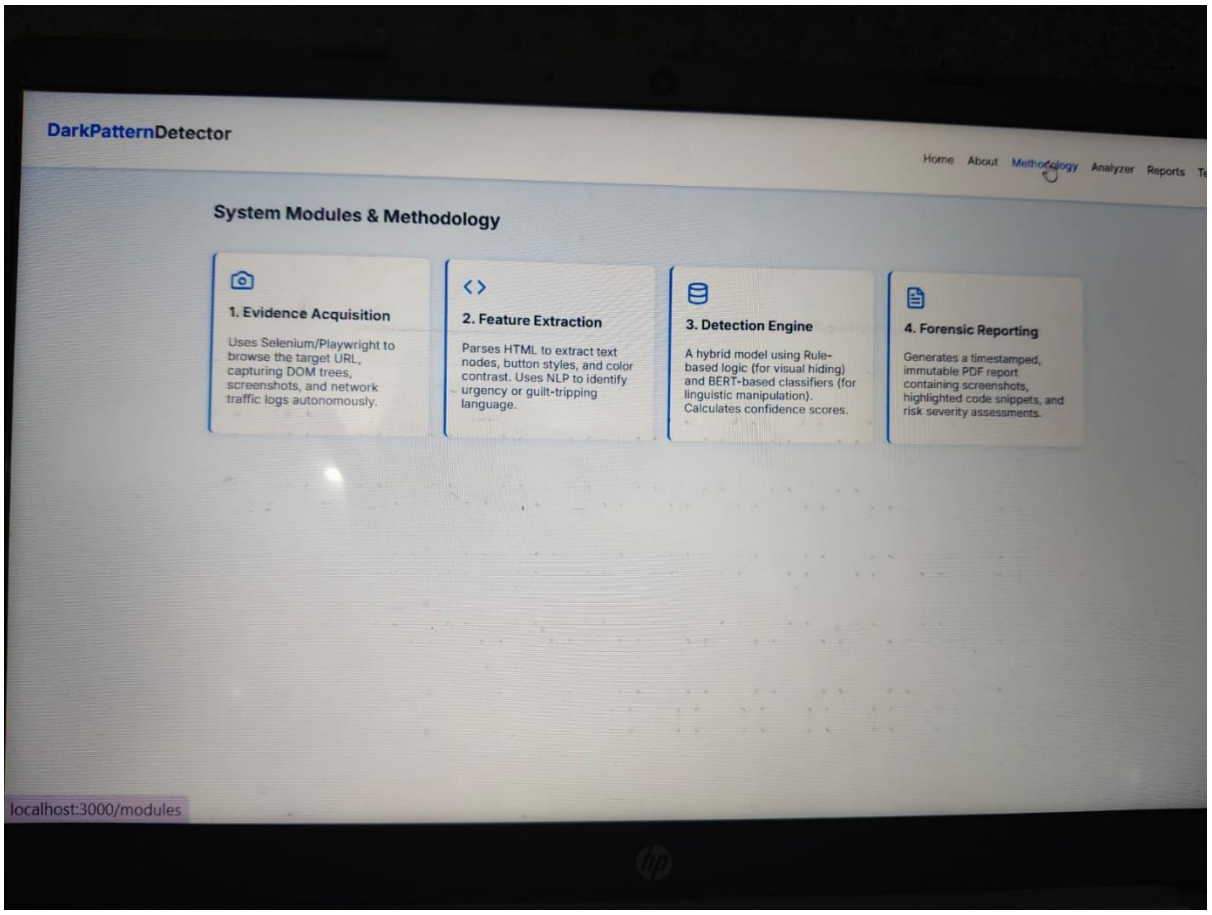

DarkPatternDetector
Home About Methodology Analyzer Reports Team
Forensic Analyzer
Enter an E-Commerce URL to scan for deceptive patterns.
https://www.flipkart.com/
Start Analysis

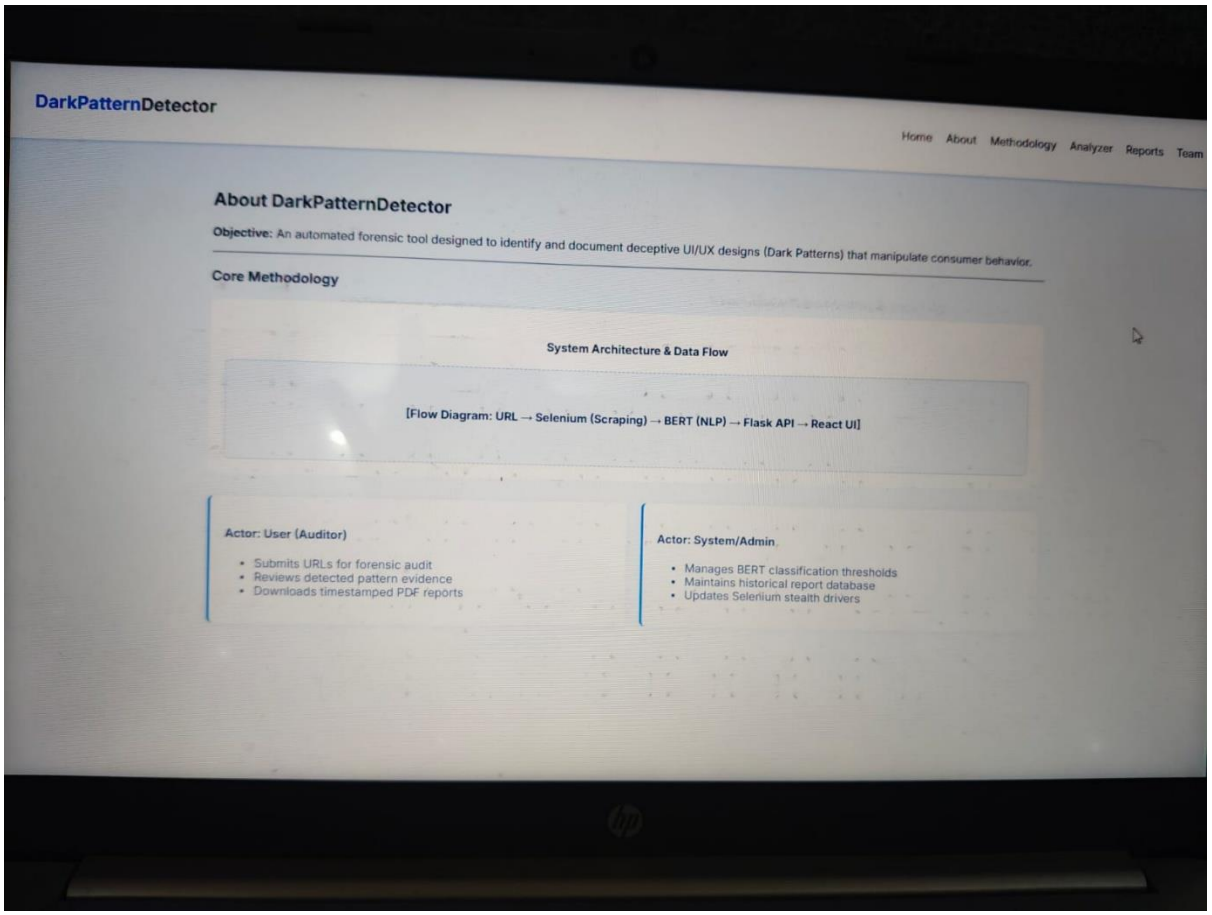

Home About Methodology Analyzer
Forensic Analyzer
Enter an E-Commerce URL to scan for deceptive patterns.
https://www.sheinindia.in
Start Analysis
Analysis Results
Risk Score: 58/100
Target: https://www.sheinindia.in
Confidence: 96%
Timestamp: 2026-03-13 23:16:18
Detected Patterns:
Fake Urgency
High Risk
Countdown timer found suggesting false deadline.
Evidence: Timer element #countdown-timer
Hidden Costs
High Risk
Unexpected fees added at checkout.
Evidence: +$12.99 Service Fee at step 4
Visual Evidence:
Evidence
Download Forensic Report (PDF)

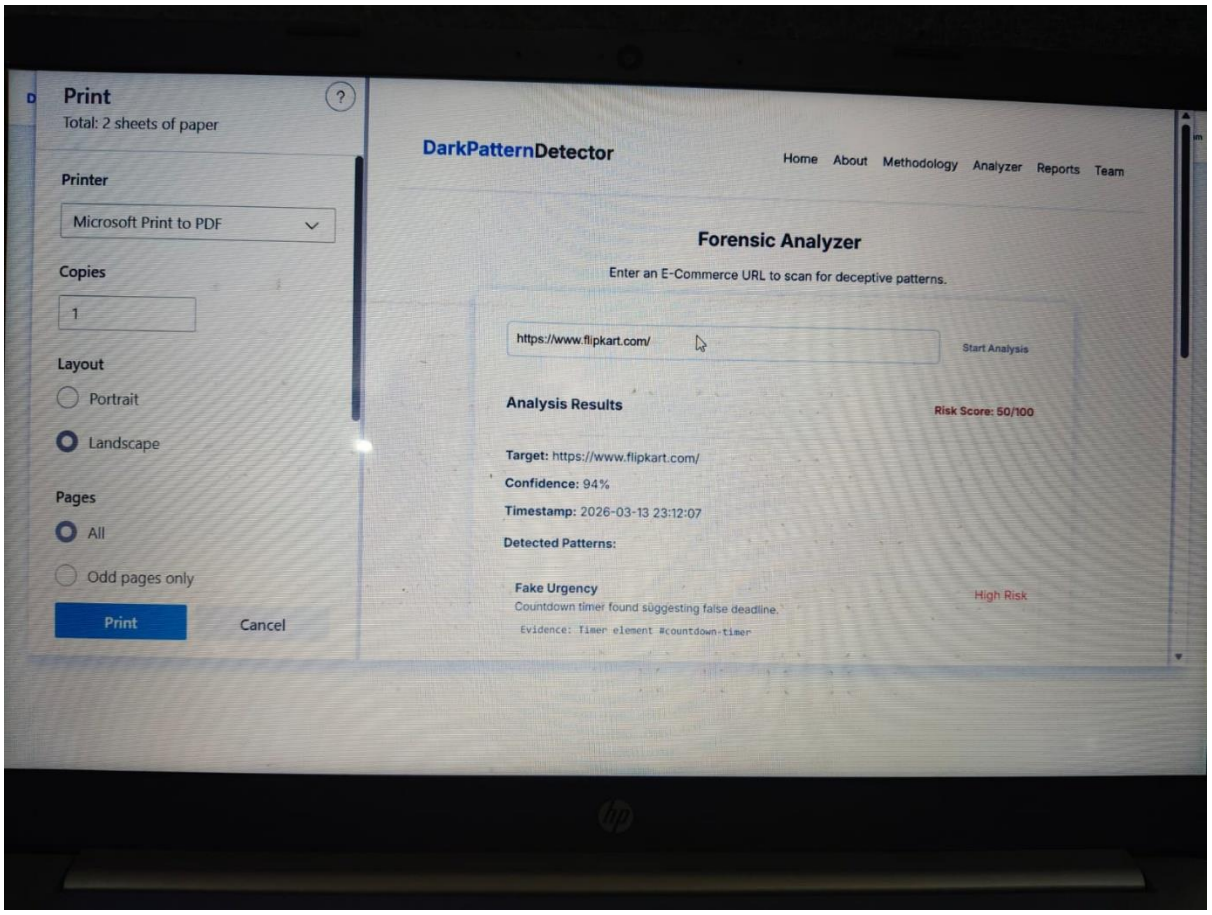

DarkPatternDetector
Home About Methodology Analyzer Reports
Project Team
Sakshi Kadadekar
Project Member
Soumya Pandey
Project Member
Veenu Sharma
Project Member
Vedika Bhaskarwar
Project Member
Mrs. Arpita Meet Vaidya
Project Supervisor
Contact Us
Your Name
Your Email
Message

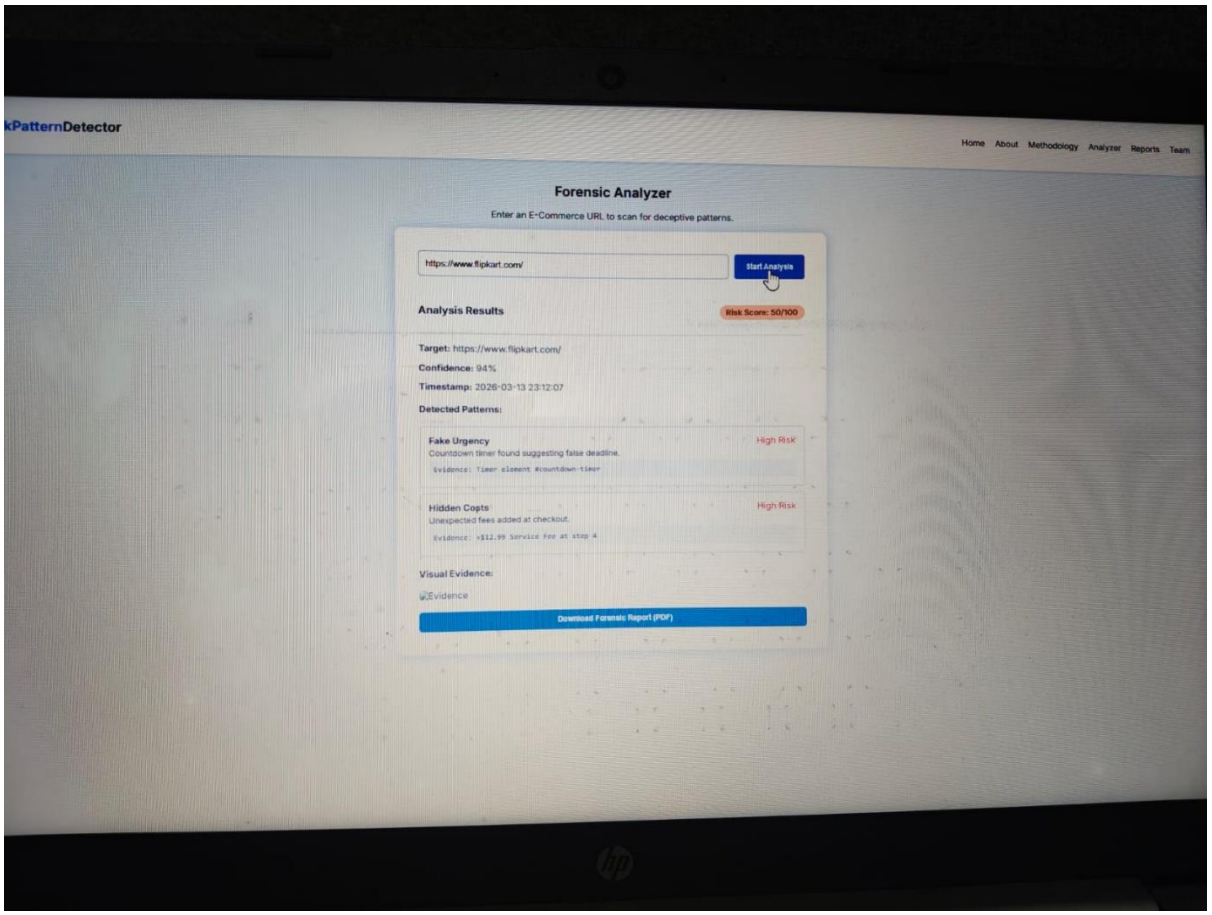

DarkPatternDetector
Home About Methodology Analyzer Reports Team
Analysis History Reports
Timestamp
Website URL
Risk Score
Patterns Found
Action
2026-03-13 23:10:11
https://www.flipkart.com/
49
3
View PDF
2026-03-13 23:12:07
https://www.flipkart.com/
50
2
View PDF
alhost:3000/reports

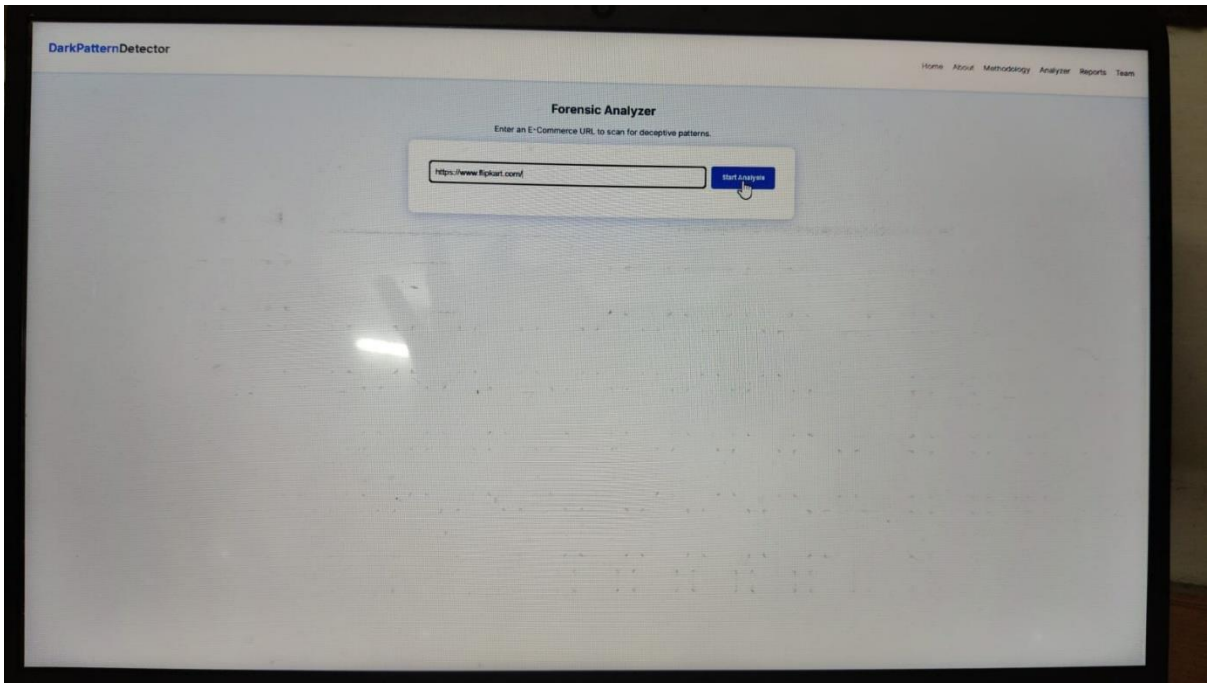

DarkPatternDetector
Home About Methodology Analyzer Reports Team
Forensic Analyzer
Enter an E-Commerce URL to scan for deceptive patterns.
https://example-ecommerce.com
Start Analysis
host:3000/analyzer


DarkPatternDetector
Home
About
Methodology
Analyzer
Reports
System Modules & Methodology
1. Evidence Acquisition
Uses Selenium/Playwright to browse the target URL, capturing DOM trees, screenshots, and network traffic logs autonomously.
2. Feature Extraction
Parses HTML to extract text nodes, button styles, and color contrast. Uses NLP to identify urgency or guilt-tripping language.
3. Detection Engine
A hybrid model using Rule-based logic (for visual hiding) and BERT-based classifiers (for linguistic manipulation). Calculates confidence scores.
4. Forensic Reporting
Generates a timestamped, immutable PDF report containing screenshots, highlighted code snippets, and risk severity assessments.
localhost:3000/modules


DarkPatternDetector
Home
About
Methodology
Analyzer
Reports
Team
About DarkPatternDetector
Objective: An automated forensic tool designed to identify and document deceptive UI/UX designs (Dark Patterns) that manipulate consumer behavior.
Core Methodology
System Architecture & Data Flow
[Flow Diagram: URL → Selenium (Scraping) → BERT (NLP) → Flask API → React UI]
Actor: User (Auditor)
Submits URLs for forensic audit
Reviews detected pattern evidence
Downloads timestamped PDF reports
Actor: System/Admin
Manages BERT classification thresholds
Maintains historical report database
Updates Selenium stealth drivers


Print
Total: 2 sheets of paper
Printer
Microsoft Print to PDF
Copies
1
Layout
Portrait
Landscape
Pages
All
Odd pages only
Print
Cancel
DarkPatternDetector
Home About Methodology Analyzer Reports Team
Forensic Analyzer
Enter an E-Commerce URL to scan for deceptive patterns.
https://www.flipkart.com/
Start Analysis
Analysis Results
Risk Score: 50/100
Target: https://www.flipkart.com/
Confidence: 94%
Timestamp: 2026-03-13 23:12:07
Detected Patterns:
Fake Urgency
Countdown timer found suggesting false deadline.
Evidence: Timer element #countdown-timer
High Risk


kPatternDetector
Home About Methodology Analyzer Reports Team
Forensic Analyzer
Enter an E-Commerce URL to scan for deceptive patterns.
https://www.flipkart.com/
Start Analysis
Analysis Results
Risk Score: 50/100
Target: https://www.flipkart.com/
Confidence: 94%
Timestamp: 2026-03-13 23:12:07
Detected Patterns:
Fake Urgency
Countdown timer found suggesting false deadline.
Evidence: Timer element #countdown-timer
High Risk
Hidden Costs
Unexpected fees added at checkout.
Evidence: +$12.99 Service Fee at step 4
High Risk
Visual Evidence:
Evidence
Download Forensic Report (PDF)


DarkPatternDetector
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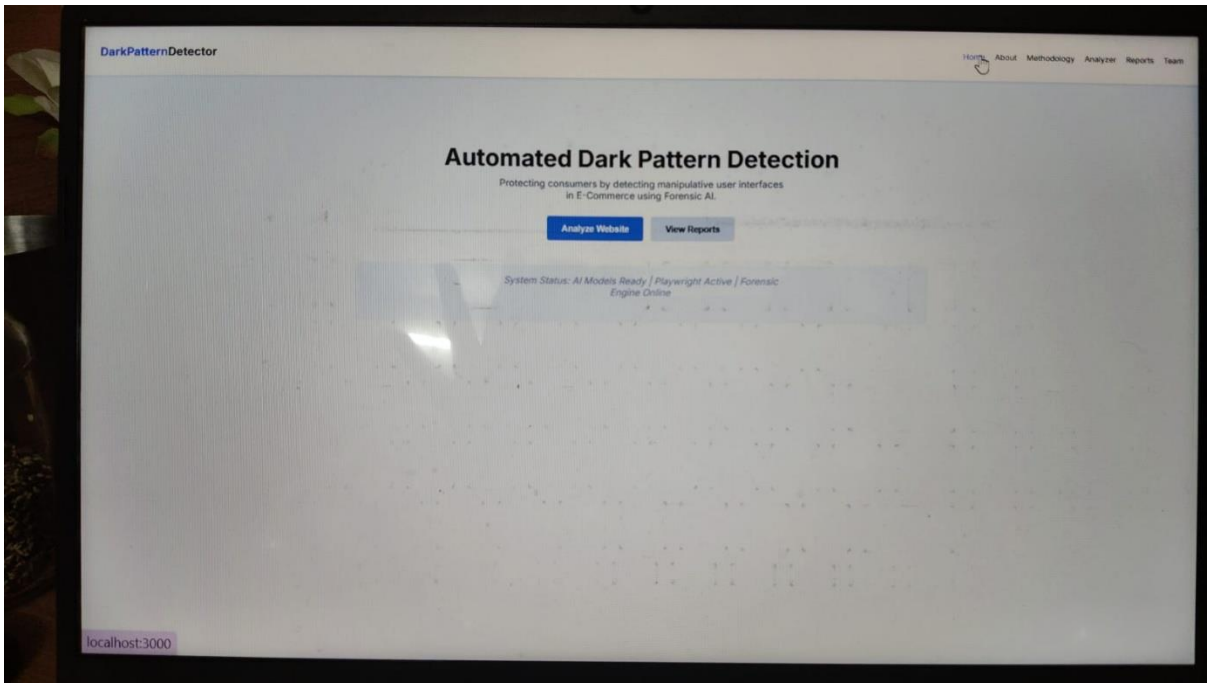
https://www.flipkart.com/
Start Analysis


DarkPatternDetector
Home About Methodology Analyzer Reports Team
Automated Dark Pattern Detection
Protecting consumers by detecting manipulative user interfaces in E-Commerce using Forensic AI.
Analyze Website
View Reports
System Status: AI Models Ready | Playwright Active | Forensic Engine Online
localhost:3000

CHAPTER VII

Research Path, Future Scope and conclusions

A novel research component is a formal **Deception Ontology** a structured model describing dark patterns mathematically. Steps include:

1. Definitions:
Formalize each pattern (e.g. **Fake Urgency** = presence of a countdown whose expiration has no effect). We can define feature functions (e.g. visual salience, choice difficulty).

2.Feature Scoring:
Develop quantitative metrics (e.g. how many pixels does the "cancel" button occupy vs. "accept"?) to score deception.

3.Probabilistic Model:
Using collected data (and possibly user response logs), build a Bayesian model relating UI features to user choices (e.g. if countdown=Yes, purchase probability=75%).

4.Casual Inference:
If A/B or time-series data is available (when a pattern was removed/added), apply causal inference (do-calculus) to estimate effect sizes of patterns.

5. FUTURE SCOPE :

Future work includes generating synthetic dark patterns: e.g., a GAN that learns to create plausible but novel deceptive UIs, to challenge the model. Also, cross-dataset benchmarks (e.g. test on unrelated web domains) and participating in any emerging dark-pattern detection challenges. Human-subject studies and legal expert validations (as mentioned above) will ground the ontology in practice.

- **Roadmap, Effort, and Risks:**

MileStones and Timeline

Milestone	Duration	Effort (person-weeks)
1. Literature Review & Planning	2 weeks	2
2. Crawler & Data Collection Prototype	4 weeks	4
3. Annotation & Dataset Assembly	6 weeks	4
4. Model Prototyping (ViT + CNN baselines)	8 weeks	2
5. Extension & API Implementation	6 weeks	1
6. Explainability & Dashboard UI	4 weeks	2
7. Evaluation (metrics & user study)	6 weeks	4
8. Deception Ontology & Write-up	4 weeks	1
9. Deployment & Finalization	4 weeks	1
Total	**44 weeks**	-

Resources:
We anticipate needing: 1–2 GPU-equipped servers (e.g. NVIDIA A100 or RTX 3090) for training; cloud instances for deployment; 3–5 annotators (graduate researchers) for labeling; and common web dev tools (Node.js, Python, Docker).

Risks & Mitigations:

Data Bias:
If training data over-represents one site type, the model may fail on others. We will ensure diverse sources and do k-fold splits by domain.

False Positives:
Overzealous detection could flag benign UI. We mitigate by conservative thresholds and manual review of borderline cases.

Adversarial Adaptation:
Sites may try to evade. We will periodically retrain with new samples and use robust features (structural as well as visual).

Legal/Ethical:
Unintended liability in labeling a design as "dark pattern." We will clearly label output as "potential" issues and allow human override. Ethical oversight (IRB, fairness audits) will be maintained throughout.

Ethical Checklist:

1. Transparency: Document how detections are made; allow site owners to request explanations.

2. Consent: Users of the extension must opt-in explicitly to any data sharing.

3. Privacy: Do not collect or store user-identifying data.

4. Fairness: Ensure no cultural/language bias (test on multilingual sites).

5. Security: Follow best practices for extension security (content security policy, minimal permissions).

CHAPTER - VIII
REFERENCES

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- FTC Bureau of Consumer Protection, *"Bringing Dark Patterns to Light"* (Staff Report, Sept. 2022)[2][3].
- European Data Protection Board, *Guidelines 3/2022 on Dark Patterns in Social Media* (Mar. 2022)[12][13].
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- Lit. surveys on dark patterns (e.g. Gray *et al.*, TOCHI 2018) and web UI datasets (RICO, WebSight).
- **Datasets/Tools**: RICO app screenshots (CVPR 2018), web crawling frameworks (Selenium, Playwright).

These sources (and others cited above) provide the empirical and theoretical foundation for this project. All algorithmic and design choices will be justified with references to these seminal works and legal texts.

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